

A8 — Interaction Distribution

This is where we make sure the HQ is not only architecturally meaningful, but **playable**.

The central question is:

What does the learner physically do in each room, and why does that interaction belong in that room?

The biggest risk now is turning the whole Headquarters into:

```
walk to monitor  
↓  
open UI  
↓  
read  
↓  
click  
↓  
next room
```

We need a mixture of:

```
PHYSICAL OBJECTS  
+  
ROOM-SCALE INTERACTIONS  
+  
TERMINALS  
+  
CHARACTERS  
+  
MOVEMENT  
+  
SEQUENCES  
+  
EVIDENCE  
+  
SIMULATION
```

A8.1 — Five interaction families

I would standardise the whole game around five families.

Family	What the learner does	Example
World interaction	Manipulates something physically in the 3D world	Pick up USB, use access reader
Spatial interaction	Moves/positions/organises objects	Build timeline, arrange evidence
Workbench interaction	Configures a system or process	Crypto bench, QKD bench
Operational interface	Uses a terminal/display for real system activity	Communications, SOC, Red Team
Character interaction	Speaks, verifies, receives/gives objects	Alice, Eve, RT operator

A good room should normally contain at least **two or three** of these types.

A8.2 — Permanent versus incident-specific interaction

Every room should have two layers:

```

PERMANENT ROOM FUNCTION
+
INCIDENT-SPECIFIC CONTENT

```

For example:

```
SOC
```

```
PERMANENT
```

- evidence board
- session viewer
- analyst consoles

```
INCIDENT 04
```

- PQ-004 timeline
- suspicious A7 session
- 172.18.44.7 evidence

```
INCIDENT 06
```

- trace graph
- candidate-system correlation
- financial/cyber evidence

The room stays recognisably SOC.

Only the case changes.

That will save us from rebuilding rooms for every incident.

A8.3 — Room-entry interaction

The room-entry popup should be **location awareness**, not a tutorial.

First entry:

```
PHANTOM Q ENGINEERS
```

```
CRYPTOGRAPHY LAB
```

```
Protection & Key Management
```

```
ACCESS
```

```
Training Authorised
```

Then fade.

On later re-entry:

```
CRYPTOGRAPHY LAB
```

small and brief.

For restricted rooms:

```
RED TEAM OPERATIONS
```

```
MISSION AUTHORISATION REQUIRED
```

The room-entry system should therefore read:

```
RoomDefinition
```

```
+
```

```
AccessState
```

```
+
```

```
CurrentIncident
```

rather than being individually scripted.

A8.4 — Entrance / Reception / Identity

This room should feel very physical.

Permanent interactions

```
ENTRY GATE  
RECEPTION DESK  
IDENTITY STATION  
TRAINING CREDENTIAL TERMINAL  
ACCESS READER  
BADGE / ID OBJECT
```

Player actions

```
Approach desk  
↓  
Register  
↓  
Create fictional training credential  
↓  
Present credential  
↓  
Complete Eve-authorised check  
↓  
Receive badge/access state  
↓  
Use access reader  
↓  
Enter HQ
```

The player should physically experience entering the organisation.

We should not reduce this to one menu.

A8.5 — Reception interaction style

The dominant interaction families are:

```
CHARACTER  
+  
WORLD  
+  
IDENTITY UI
```

Very little complex technical gameplay.

That is appropriate.

The learner is still being onboarded.

A8.6 — Central Operations

Central Operations is the most important persistent interaction room.

It needs to support multiple campaigns without becoming visually overloaded.

I would divide it internally into four functional interaction areas:

CENTRAL OPERATIONS

- |
- |— A. WORKSTATION ZONE
- |
- |— B. MAIN OPERATIONS DISPLAY
- |
- |— C. COMMUNICATION / INCIDENT INTERFACE
- |
- |— D. COMMAND / AUTHORISATION SUB-ZONE

A8.7 — Workstation 04

This is a **hero interactable**.

Not just another computer.

It persists from beginning to end.

Possible interaction states:

PROLOGUE

login
USB
files
Bob communication

INCIDENT 03

payment workflow

INCIDENT 04

payment consequence

The desk itself becomes part of the story.

That gives it emotional and spatial continuity.

A8.8 — Workstation interaction handling

When the player approaches:

```
Use Workstation 04
```

Then:

```
auto-align  
↓  
camera moves closer  
↓  
character enters use/typing pose  
↓  
workstation interface opens
```

On exit:

```
UI closes  
↓  
camera restores  
↓  
player returns to movement
```

We should avoid snapping into a completely disconnected full-screen webpage where possible.

The 3D workstation should remain visually present around the interface.

A8.9 — Main Operations Display

This should not be another computer.

It is a **room-scale status object**.

Examples:

```
PROLOGUE
normal operational status

PQ-001
incident open

I04
financial impact / affected link

I06
recovery progress

ENDING
Phantom Q restored
```

It communicates world state without requiring interaction every time.

Occasionally:

```
Review Incident Status
```

can open a more detailed view.

A8.10 — Central Operations character interactions

This is the natural space for:

```
Alice
HQ personnel
operational staff
```

Key actions:

```
Speak to Alice
Receive USB
Confirm instruction
Compare operational information
Receive mission status
```

Central Ops should therefore feel socially alive rather than machine-only.

A8.11 — Foundations room

This room should be deliberately less terminal-based.

Its gameplay should feel like **investigation through physical reasoning**.

I would create several interaction stations.

FOUNDATIONS

- |
- |— Timeline Wall
- |— Evidence Classification Table
- |— Source-Transfer-Receiver Trace
- |— CIA Assessment Station
- |— Identity Comparison Board
- |— Verification Logic Board

A8.12 — Timeline Wall

Player physically arranges event cards:

```
File 1 normal
File 2 normal
File 3 attempt 1
Bob receives nothing
retry
Bob receives altered file
credential event
```

Drag/move/order.

Not multiple-choice.

This is a spatial interaction.

A8.13 — Fact / Assumption / Unknown table

Three physical zones:

```
FACT
ASSUMPTION
UNKNOWN
```

Player takes evidence cards and places them.

Examples:


```
Bob did not receive File 3  
→ FACT
```

```
Eve changed the file  
→ ASSUMPTION
```

```
Someone read the file  
→ UNKNOWN
```

This gives us object-led learning instead of paragraphs.

A8.14 — Source–Transfer–Receiver trace

A tabletop or wall diagram:

```
ALICE  
↓  
TRANSFER  
↓  
BOB
```

The player attaches evidence to each stage.

This is much more visual than reading about integrity.

A8.15 — CIA interaction

Do not make CIA a standard quiz.

Use the actual incident evidence.

The player maps:

```
No receipt  
→ Availability
```

```
Altered copy  
→ Integrity
```

```
Whether someone read it  
→ Confidentiality in question
```

This is an evidence-assessment interaction.

A8.16 — Verification logic board

Player assembles:

```
CHECK SOURCE
↓
CHECK TRANSFER
↓
CHECK RECEIVER
↓
COMPARE
↓
MATCH?
```

This is our first algorithm-building mechanic.

We should reuse this type later.

It gives Foundations a distinctive gameplay identity:

organise, classify, trace, reason.

A8.17 — Cryptography Lab

Crypto should feel like a **technical workbench**, not a classroom.

Core permanent interactables:

```
PLAINTEXT STATION
KEY STATION
ENCRYPTION BENCH
DECRYPTION / VERIFY BENCH
KEY-TRANSFER COMPARISON AREA
ASYMMETRIC WORKBENCH
SIGNATURE TEST BENCH
```

The same room evolves as the learner advances.

A8.18 — Symmetric interaction

Instead of clicking “Encrypt”, build the relationship physically:

```
PLAINTEXT OBJECT
+
```

KEY A7



ENCRYPTION MODULE



CIPHERTEXT OBJECT

The learner places/connects the correct objects.

Then the machine runs.

This is much more game-like.

A8.19 — Key A7 should be a persistent object

A7 should not exist only as text.

Represent it as a secure digital token/object in the simulation.

States:

ALICE HOLDS A7

BOB DOES NOT

later

ALICE A7 ✓

BOB A7 ✓

later

A7 COMPROMISED

RETIRED

The visual identity persists across incidents.

This makes Incident 04 much more meaningful.

A8.20 — Key-transfer comparison

Instead of text:

Option 1

normal route

Option 2

physical token

Option 3
controlled method

give three channels/objects.

Player tests each.

The system visually exposes:

risk
delay
practicality

This is an engineering-style comparison rather than a quiz.

A8.21 — Asymmetric workbench

Incident 05 should use the **same Crypto Lab**, but new equipment becomes available.

For example:

ASYMMETRIC BENCH

Public component
Key pair generator
Public/private separation
Recipient protection
Signature system
Tamper test

This makes the laboratory feel like it is growing with the learner.

A8.22 — Signature tamper interaction

Very important opportunity for physical interaction:

SIGNED COMMAND
↓
player changes one field
↓
VERIFY
↓
SIGNATURE FAILS

The learner experiences the result.

Not a paragraph explaining it.

A8.23 — Communications

Communications should feel different from Crypto.

Its purpose is:

Observe and control the path, not the content-protection mechanism.

Permanent interactables:

```
SOURCE TERMINAL
DESTINATION TERMINAL
SESSION ROUTE DISPLAY
TRAFFIC / METADATA DISPLAY
MESSAGE TRANSMISSION CONSOLE
REMOTE BOB INTERFACE
```

A8.24 — Communications room-scale interaction

A central wall/network display could show:

```
ALICE
  ↓
protected message
  ↓
network path
  ↓
BOB
```

During later incidents:

```
ALICE
  ↓
?
  ↓
BOB
```

or:

```
source
destination
time
session
protected content
```

without necessarily exposing plaintext.

This gives Communications a strong visual identity.

A8.25 — Bob interaction

Bob's presence should live primarily here.

Player interaction:

```
Contact Bob
Confirm identity
Request authentication
Verify authorisation
Send protected message
Receive response
```

Bob can appear in a communication window or stylised remote-presence panel.

He remains part of the world without pretending he is physically there.

A8.26 — SOC

SOC should be the most evidence-heavy room.

But again, not just monitors.

I would distribute interactions into distinct physical zones:

```
SOC
|
├─ Case Board
├─ Timeline / Correlation Wall
├─ Session Investigation Console
├─ Network Trace Table
├─ Evidence Preservation Station
└─ Incident Assessment / Containment Station
```

This gives SOC movement inside the room.

A8.27 — Case Board

Persistent board:

```
PQ-001
PQ-004
ACTIVE CASES
```

When PQ-004 becomes active, the board physically changes.

Player can:

```
Open PQ-004
Pin evidence
Review known / unknown
Track current lead
```

This becomes the investigation anchor.

A8.28 — Session investigation

Here the learner examines:

```
source
destination
time
session
key activity
```

Not raw industrial SIEM logs yet.

The UI remains simplified.

This is where:

```
172.18.44.7
```

first becomes operational evidence.

A8.29 — Network Trace Table

Incident 06 needs something more visual than a terminal.

Use a table/wall map:

```
PHANTOM Q
  ↓
172.18.44.7
  ↓
?
```

Player selects:

```
TRACE
FOLLOW
COMPARE
```

As evidence develops:

```
PHANTOM Q
  ↓
172.18.44.7 RELAY
  ↓
10.91.24.18
```

This gives the threat-hunting section a physical centre.

A8.30 — Timeline / correlation wall

For Incident 06:

```
14:32:16
10.91.24.18

14:32:17
Relay

14:32:18
PQ-004 replacement
```

Player aligns evidence.

This reuses the Foundations reasoning mechanic at a more advanced level.

That is good learning progression:

```
INCIDENT 01
simple event ordering
```


Same underlying thinking.

More sophisticated context.

A8.31 — Evidence preservation station

When evidence is found:

```
PRESERVE  
↓  
select artefacts  
↓  
create package  
↓  
sign package  
↓  
send to HQ
```

This should be a dedicated interaction point.

It makes “preserve before remediation” physically memorable.

A8.32 — SOC containment station

For actions such as:

```
Suspend A7  
Close suspicious session  
Request financial hold  
Verify restoration state
```

Use an incident-control station.

This prevents the evidence board from trying to do everything.

A8.33 — Red Team Operations

Red Team must feel like a different kind of room again.

Its verbs are:

VALIDATE
REQUEST
VERIFY
ENGAGE
SEARCH
PRESERVE
REPORT

Permanent interaction areas:

TARGET CASE TABLE
AUTHORISATION TERMINAL
ATTACK-SURFACE DISPLAY
OPERATION CONSOLE
ARTEFACT SEARCH INTERFACE
MISSION STATUS WALL

A8.34 — Target case table

The player assembles:

TARGET
10.91.24.18

PATH
172.18.44.7 → 10.91.24.18

TIME
14:32

SERVICE
Remote Management

CORRELATION
Strong

This is another spatial evidence-building mechanic.

Only when the case is complete:

REQUEST CONTROLLED ACCESS

becomes available.

A8.35 — Authorisation terminal

This should be separate from the attack console.

That separation matters.

Flow:

```
REQUEST
↓
HQ RESPONSE
↓
SIGNED AUTHORISATION
↓
VERIFY
↓
TARGET
SCOPE
ACTION
↓
MATCH?
```

Only then does the operational console unlock.

That physically reinforces:

authorisation precedes attack.

A8.36 — Attack-surface display

For Release 1:

```
WEB
REMOTE MANAGEMENT
DATA
```

Player inspects each.

The result is object/graphical rather than raw port numbers.

For example:

```
WEB
No PQ-004 relevance

DATA
```

No PQ-004 relevance

REMOTE MANAGEMENT

PQ-004 activity found

Then deeper inspection reveals:

Current interface
secured

Legacy access
exposed

Recovery console
restricted

A8.37 — Controlled-access console

Once authority exists:

INITIATE CONTROLLED ACCESS

Then the player enters an investigation interface tied to the target.

The room remains visible around them.

No “Hollywood hacker” visuals.

The UI should feel like a professional operational console.

A8.38 — Artefact search

Use categories:

FILES
SESSIONS
CONNECTION HISTORY
TRANSFER RECORDS

Known clues pinned:

PQ-004
14:32
7394

The learner searches using evidence.

Random browsing produces no meaningful result.

Focused search reveals the PQ-004 artefact.

This is a good gameplay mechanic.

A8.39 — Engineering

Even though Engineering is not yet part of the current Prologue–06 campaign, we should distribute its permanent interactions now so the room has a real purpose.

Core interaction categories:

ELECTRONICS BENCH
EMBEDDED CONTROLLER STATION
TEST / MEASUREMENT BENCH
ASSEMBLY AREA
HARDWARE DIAGNOSTICS
EQUIPMENT STORAGE
TECHNICAL SERVICE ACCESS

This is important because Phantom Q must remain an engineering environment.

A8.40 — Engineering gameplay identity

Its verbs should eventually be:

BUILD
CONNECT
MEASURE
CALIBRATE
TEST
DIAGNOSE
REPAIR

Very different from SOC or Crypto.

That difference should already influence room space.

A8.41 — Quantum Wing

Quantum needs perhaps the strongest spatial interaction design.

Instead of “use QKD computer”, we should eventually have a physical/digital hybrid bench:

```
ALICE STATION
  ↓
OPTICAL / QUANTUM PATH
  ↓
EVE INTERCEPT POSITION
  ↓
BOB STATION
```

with:

```
LASER / SOURCE
BASIS SELECTION
OPTICAL COMPONENTS
DETECTION
MEASUREMENT
SIFTING DISPLAY
QBER
```

This is where the 3D layer becomes especially valuable.

A8.42 — Quantum interaction should support both digital and physical modes

Eventually:

```
DIGITAL MODE
Unity simulation

PHYSICAL MODE
Arduino / real hardware
```

The same in-world bench could expose:

```
MODE
DIGITAL SIMULATION

or

PHYSICAL LAB LINK
CONNECTED
```

That is one of the strongest ways to unify the platform.

A8.43 — Advanced Compute

This room should not be another SOC.

Its future verbs could be:

```
SIMULATE  
MODEL  
COMPARE  
PROCESS  
VISUALISE  
RUN EXPERIMENT
```

Potential uses:

```
quantum threat simulation  
advanced cryptographic experiments  
network models  
digital twins  
large-scale security simulation
```

Its interaction design should therefore be more simulation-oriented than operational.

A8.44 — Secure Core

We should deliberately keep Secure Core interaction undefined for now.

That is actually the correct design choice.

We know:

```
highest trust  
high-value systems  
restricted access
```

But we do not yet have a Release 1 gameplay requirement that justifies inventing its consoles.

So:

```
SECURE CORE  
ARCHITECTURAL PLACEHOLDER  
+  
ACCESS SYSTEM
```

but no fake gameplay.

That is disciplined design.

A8.45 — Distribution of interaction density

Not every room should contain 20 clickable objects.

I recommend three density levels.

Low density

Reception
corridors
Secure Core threshold

Few obvious interactions.

Medium density

Central Ops
Communications
Engineering

Several useful stations with open circulation.

High density

Foundations
Crypto
SOC
Red Team
Quantum

These are dedicated gameplay spaces.

But even there, only relevant objects should be interactable.

A8.46 — Object highlighting

We should avoid the common game problem where everything glows.

Use three stages:

```
NORMAL
no highlight

NEARBY
subtle outline / indicator

ACTIVE OBJECTIVE
clearer Phantom Q marker
```

A chair should not glow because it exists.

An evidence card should glow only when it matters.

A8.47 — Interaction prompts

We should maintain explicit action language.

Examples:

```
Use Workstation 04
Speak to Alice
Inspect USB
Open Evidence Board
Arrange Timeline
Encrypt Message
Compare Key Transfer
Contact Bob
Trace Connection
Correlate Evidence
Verify HQ Signature
Request Controlled Access
Preserve Evidence
Request Hold
```

That reduces unnecessary reading because the verb itself explains the action.

A8.48 — Interaction range

Different objects need different interaction behaviour.

Small object

```
USB  
card  
key token  
  
range:  
~1.2-1.5 m
```

Workstation

```
~1.6-1.8 m  
+  
auto-align
```

Large board

```
~2.0 m
```

Character

```
~1.5-2.0 m
```

The player should never need pixel-perfect positioning.

A8.49 — Interaction markers

Every important workstation should have:

```
PLAYER_MARKER  
NPC_MARKER if needed  
CAMERA_MARKER  
LOOK_TARGET
```

For example:

```
SOC_TRACE_TABLE  
|  
├─ PlayerMarker  
├─ AnalystMarker  
├─ InteractionCamera  
└─ DisplayFocus
```

This will make animation and camera transitions much easier.

A8.50 — Interaction locking

A physically present object is not necessarily usable yet.

Example:

```
ASYMMETRIC WORKBENCH
```

```
Incident 02:
```

```
Visible
```

```
Unavailable
```

```
Incident 05:
```

```
Active
```

That creates progression inside the same room.

Likewise:

```
RED TEAM ATTACK CONSOLE
```

```
Before verified HQ authority:
```

```
LOCKED
```

```
After:
```

```
ACTIVE
```

The world itself changes with capability.

A8.51 — Permanent versus temporary interactables

I would categorise objects internally as:

```
PERMANENT
```

```
Exists every time room loads
```

```
CONTEXTUAL
```

```
Exists permanently but activates only when relevant
```

```
MISSION
```

```
Spawned/loaded specifically for an incident
```

Example:

```
SOC CASE BOARD
PERMANENT

PQ-004 CASE
MISSION → then EVIDENCE/PERSISTENT

Trace Table
PERMANENT

172.18.44.7 node
MISSION/EVIDENCE
```

This will make the Unity scene much easier to manage.

A8.52 — Evidence persistence

When the learner discovers something, it should remain available later.

Example:

```
PQ-004
|
├─ Alice original
├─ Bob transaction
├─ 172.18.44.7
├─ relay classification
├─ 10.91.24.18
├─ artefact
└─ 7394 → 1058
```

The case board progressively fills.

This gives the learner a visible sense of investigation accumulation.

A8.53 — Reusing mechanics without making rooms feel repetitive

We should reuse **logic**, not appearance.

Example:

Foundations

Arrange timeline cards

SOC

Correlate network-event timeline

Same underlying mechanic.

Different presentation and complexity.

Similarly:

Foundations

classify Fact / Assumption / Unknown

Red Team

classify Evidence / Lead / Attribution

That produces learning progression and reduces development cost.

A8.54 — Interaction complexity progression

The campaign should gradually move from simple to combined interactions.

PROLOGUE

use / inspect / confirm

I01

arrange / classify / compare

I02

configure / transform

I03

verify / execute workflow

I04

compare / investigate / contain

I05

build / verify / authenticate

I06

trace / correlate / authorise / engage / preserve / recover

This is very strong game progression.

The learner's **verbs become more sophisticated** as their capability grows.

A8.55 — Avoid excessive instructional popups

We should use a hierarchy:

ROOM INTRODUCTION

only first time

OBJECTIVE

short

CONTEXT PROMPT

verb

OPTIONAL HELP

available on demand

DETAILED EXPLANATION

only when requested / necessary

Do not present:

"Welcome to Cryptography Lab.

Cryptography is the science of..."

every time they walk in.

The room should teach through interaction.

A8.56 — Help character / assistance

Our help-desk guidance concept can fit here.

Instead of permanently filling the screen with text:

HELP

opens a character-assisted hint.

For example:

```
"I can see you have ciphertext,  
but Bob does not have A7 yet.  
  
Check how the key is being delivered."
```

The hint responds to current state.

That is much more useful than generic tutorials.

A8.57 — Interaction failure should remain contextual

We maintain the rule already established.

Examples:

Wrong timeline order

```
SEQUENCE CONFLICT
```

not:

```
WRONG ANSWER
```

Attempt attack before authority

```
AUTHORISATION REQUIRED
```

Wrong service

```
PQ-004 MATCHES: 0
```

Recover before containment

```
FUNDS REMAIN MOBILE
```

The world tells the learner **why** the action fails.

A8.58 — Input handling

All interactions should ultimately map to a small input vocabulary.

Desktop:

```
Move
Interact
Back
Inspect
Optional shortcut
```

Controller:

same conceptual actions.

Mobile:

```
virtual movement
context interaction
touch UI
```

We should not create 20 keyboard keys because Incident 06 has 20 actions.

The action changes contextually.

A8.59 — Central interaction architecture in Unity

Conceptually:

```
PLAYER
↓
InteractionDetector
↓
IInteractable
↓
InteractionContext
↓
CanInteract?
↓
Prompt
↓
Execute
↓
Mission Event
```

Then every room plugs into the same framework.

A workstation, character and USB can all implement the same base interaction contract.

A8.60 — Room-definition architecture

Each room should eventually hold something conceptually like:

```
RoomDefinition
|
├─ roomId
├─ displayName
├─ subtitle
├─ requiredAccess
├─ firstEntryMessage
├─ reentryMessage
├─ permanentInteractables
├─ missionInteractables
├─ cameraProfile
├─ ambientProfile
└─ music/audioProfile
```

This directly supports the popup system you raised earlier.

A8.61 — Interaction map for Release 1 rooms

Room	Primary interaction identity
Reception	Register / identify / authenticate / enter
Central Ops	Operate / communicate / receive / compare / restore
Foundations	Arrange / classify / trace / reason
Crypto	Build / protect / transform / verify
Communications	Send / receive / observe path / inspect metadata
SOC	Investigate / correlate / trace / preserve / contain
Red Team	Validate / request / verify / engage / search
Engineering	Build / connect / measure / test
Quantum	Configure / align / transmit / detect / analyse

Room	Primary interaction identity
Advanced Compute	Simulate / model / process
Secure Core	Deliberately undefined for current release

This is probably the most important A8 summary.

Every room now has **different verbs**.

A8.62 — Does Workstation 04 become overloaded?

This is something A8 exposes.

Potentially yes.

We must not make Workstation 04 do:

```
files
crypto
SOC
Red Team
QKD
all mission interfaces
everything
```

That would undermine the rooms.

So Workstation 04 should remain:

```
PERSONAL / OPERATIONAL WORKSTATION
```

It handles:

```
messages
files
basic operations
mission status
normal work
payment interaction
final restoration confirmation
```

Specialist operations belong in specialist rooms.

That is an important lock.

A8.63 — Does Communications need physical messaging terminals and network display?

Yes.

A8 confirms that Communications cannot simply be a corridor between Crypto and SOC.

It needs a distinct gameplay footprint.

At minimum:

```
BOB / ENDPOINT TERMINAL  
TRANSMISSION CONSOLE  
NETWORK PATH WALL  
METADATA / SESSION DISPLAY
```

So A4's 14 × 10 m size remains justified.

A8.64 — Does SOC need more room?

A8 shows that SOC must contain:

```
case board  
timeline wall  
trace table  
session console  
preservation station  
containment/control station  
staff workstations
```

Our current 16 × 12 m footprint is probably still adequate for a stylised environment, but it should remain an **open operational room**, not be overfilled.

We should test this carefully in greybox.

I would not enlarge it yet.

A8.65 — Does Red Team need 16 × 12 m?

Yes.

A8 validates it.

Because we need:

```
target case  
authorisation  
attack surface  
operation console  
artefact search  
mission display  
staff
```

while maintaining clear circulation.

So the A4 dimension remains defensible.

A8.66 — Does Foundations need 14 × 12 m?

Yes.

Its interaction style uses physical boards and tables.

A smaller classroom-like room would constrain gameplay.

This confirms its footprint.

A8.67 — Does Crypto need 14 × 12 m?

Yes.

Especially because the same room supports:

```
symmetric  
key-transfer  
asymmetric  
signature
```

over time.

We need enough space for old equipment to remain visible while new capability activates.

A8.68 — Loop against A1–A7

A1 — Meaning

Passes. Rooms now exist because they perform distinct organisational functions.

A2 — Hierarchy

Passes. Interaction complexity becomes more specialised deeper into HQ.

A3 — Adjacency

Passes. Directly connected departments also have complementary interaction roles.

A4 — Floor plan

Passes. No room relocation needed; SOC density should be tested in greybox.

A5 — Campaign routes

Passes. Movement occurs when the interaction mode genuinely changes.

A6 — Character routes

Passes. Alice/Bob/Eve/RT interactions align with their physical roles.

A7 — Clearance

Passes. Objects can be visible, locked, limited, mission-enabled or fully usable independently of room access.

A8 — Canonical lock

The central interaction philosophy is now:

```
ROOM
defines operational purpose

OBJECT
defines available action

MISSION
defines relevance

CLEARANCE
defines permission

PLAYER
acts physically where possible

UI
supports the action
rather than replacing the world
```

And most importantly:

Each Phantom Q department now has its own action vocabulary.

RECEPTION

identify

CENTRAL OPS

operate

FOUNDATIONS

reason

CRYPTO

protect

COMMUNICATIONS

transmit

SOC

investigate

RED TEAM

engage

ENGINEERING

build

QUANTUM

measure

ADVANCED COMPUTE

simulate

That is what will stop Phantom Q from becoming a sequence of terminals with different wallpapers.

The next loop, **A9 — Camera and Handling**, should now take this exact interaction distribution and determine how the camera behaves when the player is **walking, entering rooms, speaking to characters, manipulating a table, using a workstation, operating a Crypto bench, investigating SOC evidence and entering Red Team mode**. That is where we make the whole system feel consistent in the player's hands.